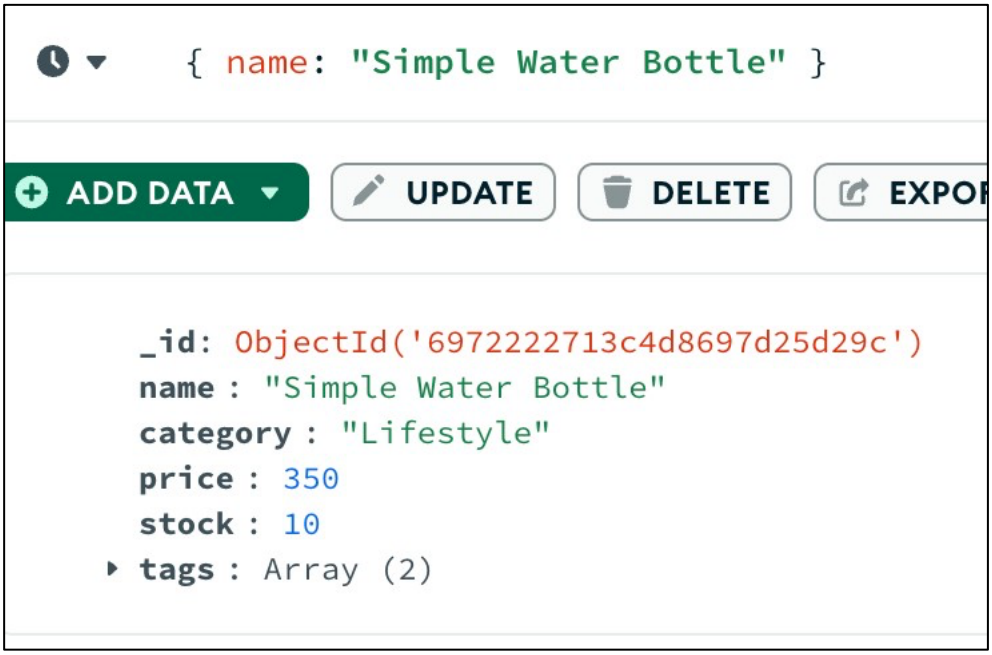
Task2 : NoSQL Data Models

Task A (Insert) — Product Collection

Write the MongoDB command to insert a new product document.

```
db.products.insertOne([{
  name: "Simple Water Bottle",        category:
"Lifestyle",
  price: 350,   stock: 10,
  tags: ["Healthy","student utilities",]
   }]
);
```



```
db.products.insertOne({ name: "Onida Ac", category: "Electronics", price: 50000, tags: "Air conditioners, Lifestyles" })
```

```
_id: ObjectId('697239b9c3edee440f521908')
name : "Onida Ac"
category : "Electronics"
price : 50000
tags : "Air conditioners, Lifestyles"
```

```
db.products.insertOne({ name: "Onida Ac", category: "Electronics", price: 50000, tags: "Air conditioners, Lifestyles" })
```

```
_id: ObjectId('69723a04c3edee440f521909')
name : "Goggles"
category : "Accesories"
price : 500
tags : Array (2)
```

Explanation:
These queries insert new product documents into the products collection. Each document includes fields like name, category, price, stock, and tags. The first query uses insertOne with an array (though typically insertOne expects a single document), while the others insert individual product documents. This adds new products to the database for later retrieval and management.

Task B (Find) — Product Collection

Write the MongoDB command to find all products where the price is greater than 2000.

```
db.products.find(
 { "price": { "$gt": 2000 } },
 { "name": 1, "_id": 0 }
)
```

```
< {
    name: 'Bluetooth Headphones'
  }
  {
    name: 'Office Chair'
  }
  {
    name: 'Bluetooth Headphones'
  }
  {
    name: 'Office Chair'
  }
  {
    name: 'Study Table'
  }
  {
    name: 'Dumbbell Set'
  }
```

- Query to display names of products with price less than 2000

```
db.products.find(
 {"price": {"$lt" : 2000} },
 { "name": 1, "_id":0 })
```

```
< {
    name: 'Wireless Mouse'
  }
  {
    name: 'Notebook Pack'
  }
  {
    name: 'Smart Water Bottle'
  }
  {
    name: 'Wireless Mouse'
  }
  {
    name: 'Laptop Stand'
  }
  {
    name: 'Notebook Set'
  }
  {
    name: 'Ball Pen Pack'
  }
```

- Extract stationery products

```
db.products.find(  {"category":
"Stationery" },
 { "name": 1, "_id":0 }
)
```

```
< {
    name: 'Notebook Pack'
  }
  {
    name: 'Notebook Set'
  }
  {
    name: 'Ball Pen Pack'
  }
```

Explanation:

These queries retrieve products based on specific conditions:

- The first query finds all products with price > 2000 and returns only the name field.
- The second query finds products with price < 2000 and returns the name field.
- The third query extracts products where category is "Stationery" and returns only the name. These queries filter and project data for easier reporting or analysis.

Task C (Projection) — Product Collection
Write the MongoDB command to find all product documents but return only the product name field (exclude the _id).

```
db.products.find(
 {},
 {"_id":0, "name":1, "stock":1 }
)
```

```
< {
    name: 'Wireless Mouse',
    stock: 120
  }
  {
    name: 'Bluetooth Headphones',
    stock: 60
  }
  {
    name: 'Office Chair',
    stock: 25
  }
  {
    name: 'Notebook Pack',
    stock: 200
```

Explanation:
This query retrieves all products but returns only the name and stock fields while excluding the _id. Projection limits the fields returned in the results, making queries more efficient and focused on the required information.